User guide

Smart Parking shows a parking lot, finds the nearest free spot for an arriving car, and guides the driver there. One spot is real: an ESP32 with two IR sensors and an LED strip. The rest of the lot is simulated. (עברית: USER_GUIDE.he.md)

What you need

- A computer with **Chrome or Edge** (Safari and Firefox cannot talk to USB)
- The ESP32 node, wired as in HARDWARE.md, with the firmware uploaded
- The site (the Vercel link, or `npm run dev` and `http://localhost:5173`)

No board? You can still try everything with the on-screen emulator, or with the board's mock mode.

Start it

1. Plug the board into the computer with a USB cable. The LED strip flashes **blue** for a moment, then shows **green** (spot free).
2. Close the Arduino IDE Serial Monitor if it is open.
3. Open the site. It starts on the **Overview** page; press **Open the live demo**, then **Connect Arduino**. Pick the board's port (it says CP2102 or USB Serial).
4. The top of the Hardware panel says **Board connected** and the two sensor bars move. Next time, the site reconnects by itself.

Use it

1. **Put a toy car in front of the entrance sensor.** The phone shows "Car detected", then "Finding nearest spot", then turn-by-turn directions, and the car drives on the lot map.
2. The car stops next to the real spot ("Pull into Spot S25"). **Move the toy car to the spot sensor.** The LEDs turn **red** and the app confirms you have parked.
3. The real-sensor tag (**HW**) moves to the next spot. Clear the spot sensor and bring the next car.

Other things to try: click a free spot to fill it, click a lane to close it for roadwork (click it again to reopen it), turn on **Live Traffic**, drag the **HW** tag to another spot, switch to the **Simulation study** page for the numbers behind the routing. A car that arrives while another one is being guided, or while no free spot can be reached, waits at the gate; the phone shows how many are waiting, and they drive in one by one.

Edge cases to try (below the map) sets up a hard situation with one click and plays it: one spot left, someone takes your spot, lot full, roadwork in an aisle, free spots with no way in, three cars at once, rush hour, and a noisy sensor reading. Each card says what to watch for.

Below the map:

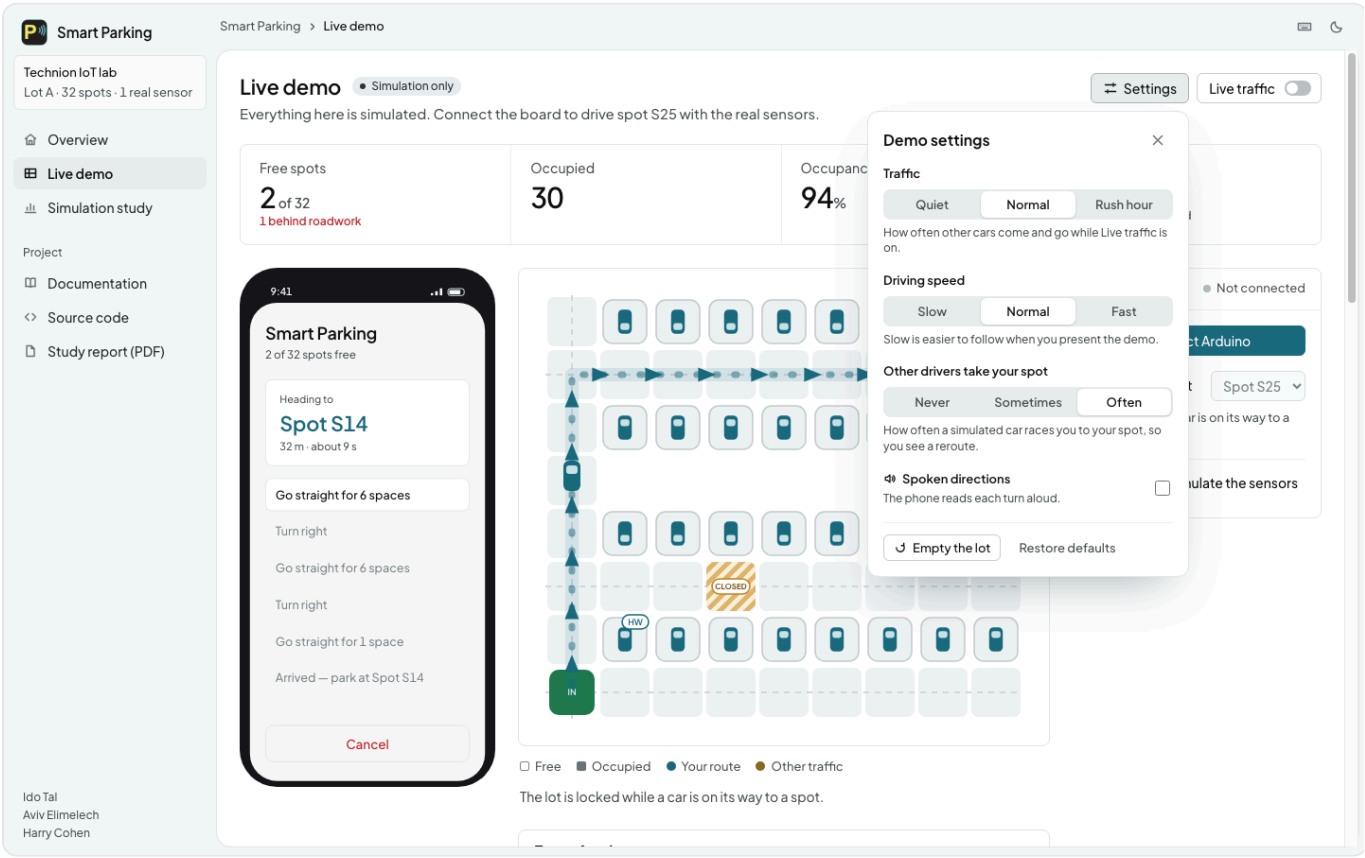
- **Live sensor readings** draws the last 30 seconds of both IR sensors with their trigger distances, and shades the moments the board decided "car" or "taken". **Pause** freezes it; **Export CSV** saves the readings.
- **This session** counts the cars guided, their average drive in metres, reroutes and sensor-confirmed parks, and plots how full the lot was over time. **Export CSV** saves the summary and the whole event feed.

The moon / sun button at the top right switches between the light and the dark theme.

Settings (next to Live traffic) changes how the demo plays. The choices are remembered in the browser:

Setting	Choices
Traffic	Quiet, Normal, Rush hour: how often other cars come and go
Driving speed	Slow (good for presenting), Normal, Fast
Other drivers take your spot	Never, Sometimes, Often: how often a simulated car races you to your spot
Spoken directions	the phone reads each turn aloud (in browsers that can speak)

Empty the lot frees every spot and opens every lane; **Restore defaults** resets the settings.



Keyboard shortcuts

Key	Does
`1` `2` `3`	Overview, Live demo, Simulation study
`F`	send a car to the gate (live demo)
`T`	live traffic on or off
`E`	empty the lot and open every lane
`D`	dark or light theme
`?`	show the list of shortcuts

What the LEDs mean

LEDs	Meaning
green	the spot is free
red	the spot is taken
blue, steady	starting up
blue, fast blink	calibrating a sensor
orange, blinking	a sensor is broken (check its wires)
last LED violet	mock mode: the data is generated, not measured
first LED blinks amber	the app stopped talking to the board (the tab was closed, the app disconnected, the laptop went to sleep). The spot colour stays on the other LEDs and the board keeps working
first LED blinks cyan	WiFi is switched on but the network is gone (the board keeps working offline)

The BOOT button on the board

Press	Does
short	mock mode on / off
hold 2 seconds	restore the default settings

Settings

In the Hardware panel:

- **Entrance trigger / Spot trigger**: how close (in cm) something must be to count as a car / a taken spot. They are saved on the board.
- **Board settings and technician view** (open it with the small arrow): **Calibrate** buttons (put the car in place first, then press), **Mock mode**, **LED brightness**, **Restore board defaults**, and **Show raw data**.

If something goes wrong

You see	Do this
"Could not open the port..."	Close the Arduino IDE Serial Monitor (or any other program using the board), press Connect again
The port is not in the list	Use a USB cable that carries data; install the CP2102 driver; replug the board
"Connected, but the board is not sending data"	Upload the sketch again, make sure the baud rate is 115200
"Board not responding"	Check the cable or press the board's reset button. Detection resumes by itself
"Lost connection to the board"	The cable came out. Plug it back in; the site reconnects
"A sensor looks broken" and orange LEDs	Check that sensor's three wires (signal, power, ground)
A car is detected too early or too late	Calibrate (Board settings) or move the trigger slider
Nothing works at all	Press **Mock mode** to confirm the site works, then check the wiring
No Connect button	Use Chrome or Edge on a computer; or use the emulator

More cases: EDGE_CASES.md.